

Artifacts are noncerebral waveforms that mimic cerebral activity. Excess artifact may hide underlying abnormalities.

When artifacts originate from within the body (organs other than the brain) they are called internal or physiologic artifacts. Those originating outside the body are called external or nonphysiologic artifacts [1].

Closely examine the bedside environment either in person or on video to identify the cause. Numerous external and internal artifacts can occur, some commonly encountered ones are described below.

Common External Artifacts

Electrode Artifact

High electrode impedance or a sudden change in impedance (electrode pop) leads to artifact in the affected channels. Often nearby alternating current sources such as electrical devices (e.g., dialysis machines, pumps, beds) will also reflect in these problem electrodes causing more artifact (60 Hz artifact).

They have a variable appearance but regardless of montage electrode artifacts are limited to the channels containing the problem electrode (no field) [2].

Significance

1. They may be spiky, periodic, or rhythmic and may mimic epileptogenic or ictal patterns.
2. Electrode artifact (e.g., 60 Hz) can cause unusual patterns in several channels, these can sometimes mimic cerebral background activity. Turning off the 60 Hz filter reveals the underlying artifact.
3. Fix or replace the electrode, turn off any nearby devices (such as pumps), or use a filter to decrease this artifact.

Figure 12.1 shows spiky appearing electrode artifact at T8 and Figure 12.2 shows “pop” artifact at P7.

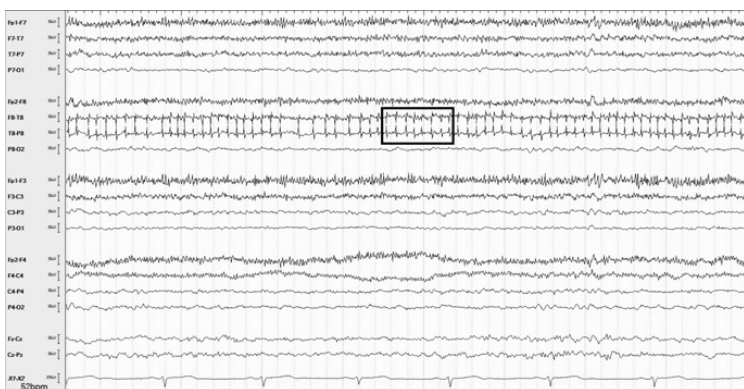


Figure 12.1 Thirty-eight-year-old man with alcohol intoxication, spiky periodic T8 electrode artifact.

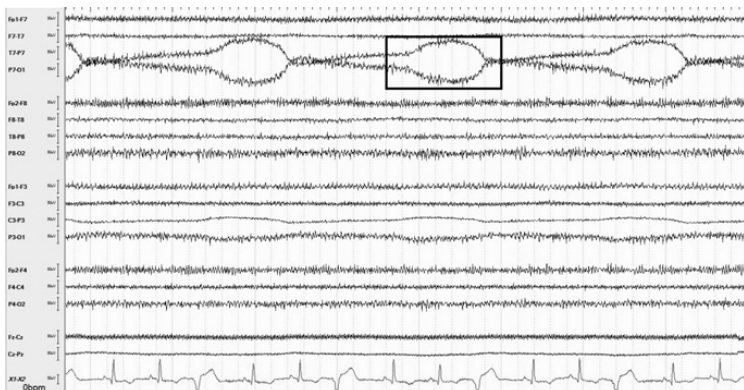


Figure 12.2 Seventy-two-year-old man in a coma, “electrode pop” artifact at P7. Note the irregular heart rate.

Sweat Artifact

This occurs due to a moist scalp. It is typically a slow sinusoidal delta pattern that can appear in a single or multiple channel [3].

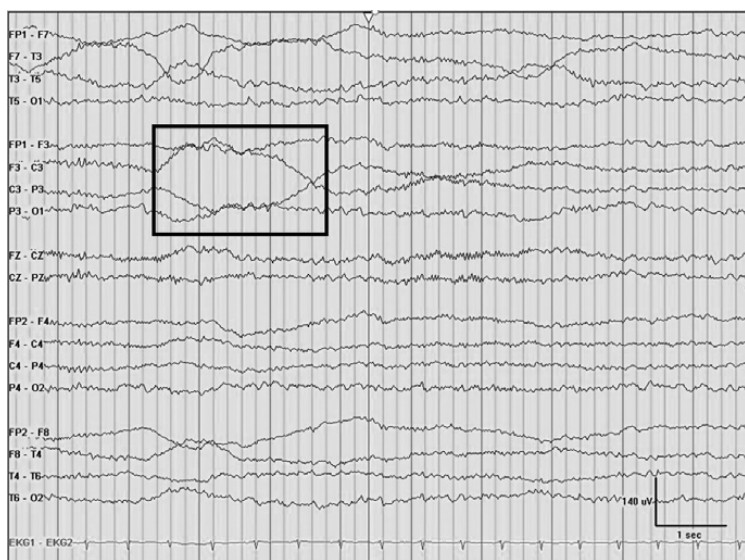


Figure 12.3 Fifty-seven-year-old man with encephalopathy, sweat artifact in multiple channels.

Significance

1. May mimic slow lateral eye movements (esp. in F7 and F8) or generalized rhythmic delta activity (GRDA).
2. Wipe the scalp, cool the patient, or use a low-frequency filter to decrease this artifact.

Figure 12.3 shows sweat artifact in multiple channels.

Ventilator Artifact

This typically leads to a regular paced slower waveform (single or burst-like). Occasionally, faster activity may result from fluid in the tubing. Commonly affects the anterior channels but variable in location and appearance. Figure 12.4 shows a periodic appearing ventilator artifact occurring every 6 s and corresponding to the set respiratory rate [3].

Significance

1. Monitoring the ventilation rate either using a separate channel (mouth electrodes) or at bedside helps identify the artifact.
2. May mimic bursts of cerebral activity in comatose patients.

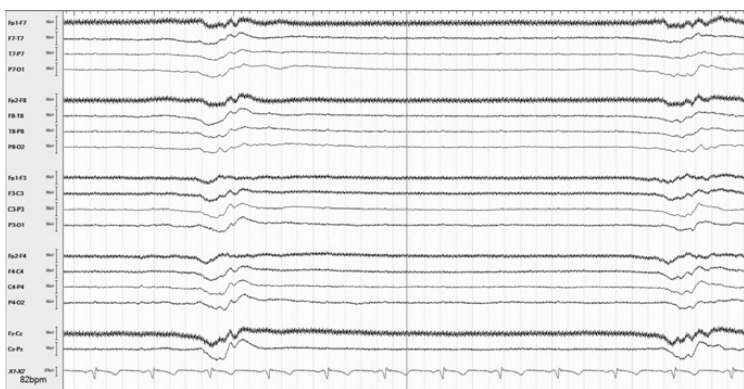


Figure 12.4 Eighty-year-old woman in a coma, slow-paced ventilator artifact.

Others

Rhythmic artifact may result from telephone ringing, bed-percussion, or devices such as Continuous Renal Replacement Therapy (CRRT), and Extracorporeal Membrane Oxygenation (ECMO).

Cardiopulmonary resuscitation, sternal rub, chest percussion, tooth brushing, and other maneuvers also result in fast rhythmic artifacts that may mimic ictal activity [3]. Figure 12.5 shows diffuse (including EKG channel) rhythmic appearing artifact from brushing teeth.

Common Internal Artifacts

The eyes, tongue, heart, and muscles are common causes of internal artifact.

Eye Artifacts

Vertical Eye Movements (Includes Eye Blinks): The eye is a dipole with the cornea charged positive and the retina charged negative. With lid closure the cornea rolls up toward the frontopolar channels (Fp1 and Fp2) due to Bell's phenomena, leading to a large downward deflection (positive is down) in those channels. Eye opening will have the opposite effect (upward deflection). Figure 12.6 shows eye blink artifact and Figure 12.7 shows vertical eye movement artifact in the Fp1 and Fp2 channels.

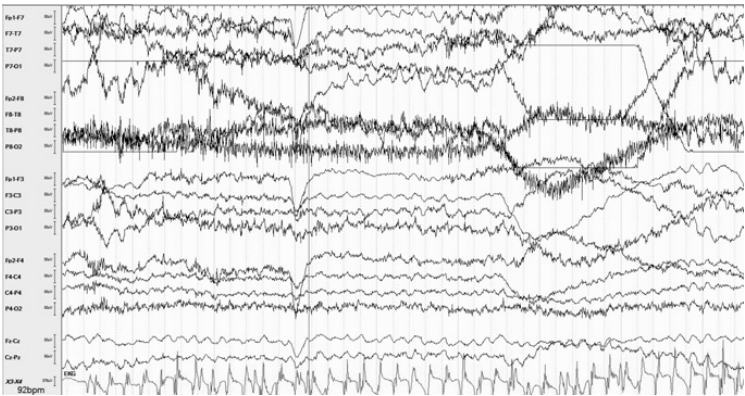


Figure 12.5 Thirty-five-year-old man with shaking spells, diffuse rhythmic toothbrushing artifact.

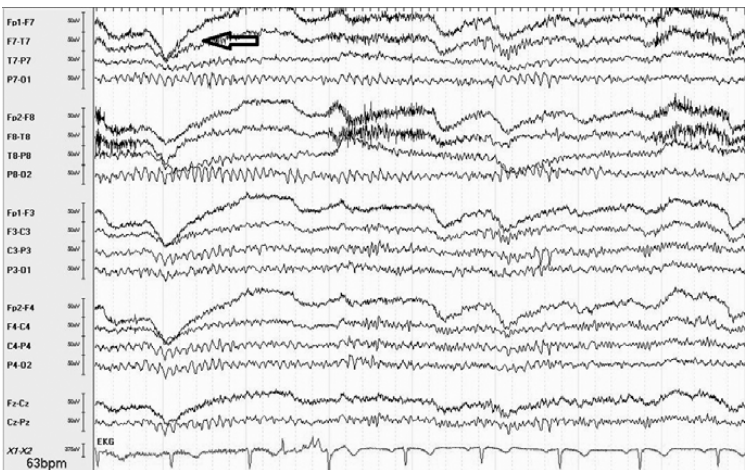


Figure 12.6 Thirty-seven-year-old woman with seizures, eye blink artifact in the frontopolar (Fp) channels.

Significance

1. Vertical nystagmus or ocular bobbing may also result in a similar artifact. Vertical nystagmus has a sharp upward deflection (corresponds to



Figure 12.7 Thirty-five-year-old man with shaking spells, vertical eye movement artifact in the frontopolar (Fp) channels.

a quick down phase) and slow downward deflection (slower return to up gaze).

2. Eye lid flutter may mimic frontally predominant generalized rhythmic delta activity (GRDA). However, unlike GRDA, ocular artifacts don't extend beyond the frontopolar derivations. Figure 12.8 shows prominent eye flutter artifact.
3. In rare circumstances inferior orbital electrodes may be used to confirm vertical eye movement artifact where the waveforms are always out of phase compared to the frontopolar channels [1,2].

Lateral Eye Movements: Similar principles apply (the cornea is positive relative to the retina), but these waveforms are seen in F7 and F8. If the eyes look to the left (F7) then the positive cornea causes a downward deflection in the F7 channel and a corresponding upward deflection in F8. When looking to the right (F8) there is an upward deflection in F7 and a downward deflection in F8. Figure 12.9 shows lateral eye movement artifact during wakefulness. Lateral rectus spikes characterized by a small artifactual spike (commonly in the F7/F8) channel may precede lateral eye movements [1,2].

Significance

1. Lateral nystagmus (e.g., due to seizures) may cause similar artifact. Figure 12.10 shows artifact produced by lateral nystagmus in the left eye (F7 channel).

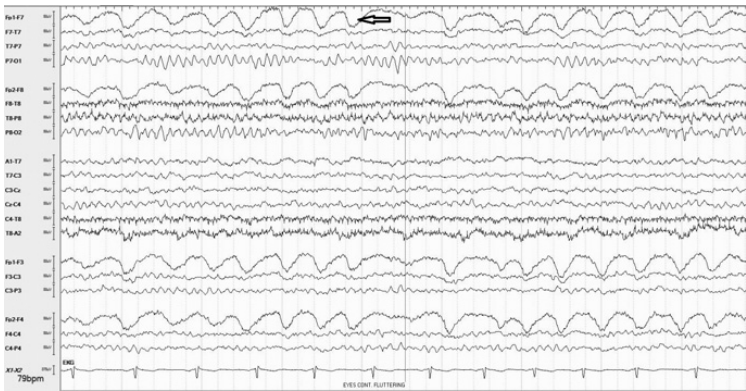


Figure 12.8 Twenty-eight-year-old woman with anxious spells, eye flutter artifact in the frontopolar (Fp) channels.

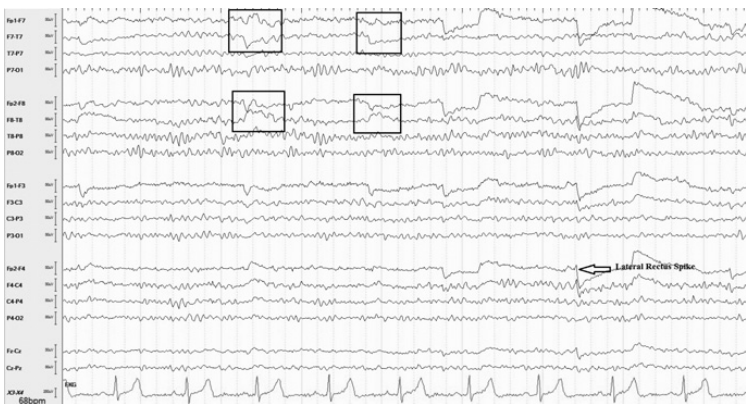


Figure 12.9 Thirty-five-year-old man with shaking spells, horizontal eye movement artifact in F7/F8 and lateral rectus spikes.

2. Rapid eye movement (REM) sleep results in lateral eye movement artifact but without eye blinks. Figure 12.11 shows lateral eye movements during REM sleep.

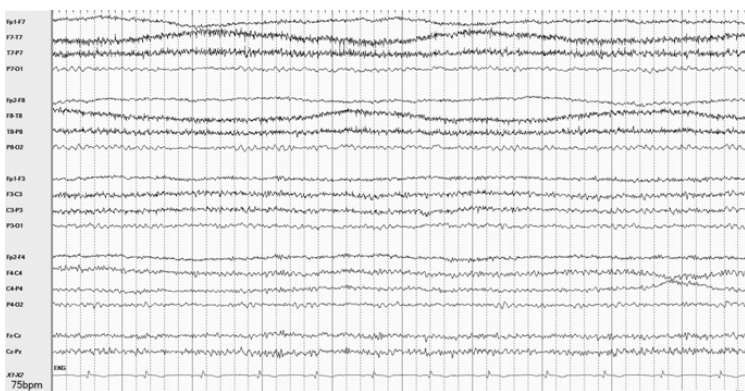


Figure 12.12 Forty-seven-year-old woman in drowsiness, slow rowing eye movement artifact at F7/F8.

Tongue (Glossopharyngeal) Artifact

The tongue is a large muscle with its tip relatively negative compared to its base. Tongue movements (such as eating, swallowing, or hyperventilation). This appears as a burst of frontally predominant high-amplitude delta slowing. Typically, there is some superimposed muscle artifact seen. It is best appreciated in the frontopolar channels but unlike ocular artifacts has a wide field that may extend to the posterior channels [2]. Figure 12.13 shows glossopharyngeal artifact while talking.

Significance

1. It may be confused with frontally predominant generalized rhythmic delta activity (GRDA).
2. The artifact may be reproduced by repeating lingual sounds such as “la-la-la” or confirmed with placement of submental electrodes.
3. Sipping from a straw may also produce a similar rhythmic artifact of delta frequency that mimics GRDA. Figure 12.14 shows a generalized rhythmic artifact produced while using a drinking straw.

Cardiac Artifact

Electrocardiogram (EKG Artifact): Regular cardiac contractions can result in a periodic artifact that corresponds with the QRS complex on the electrocardiogram channel. It is most prominent in the ear referential montages.

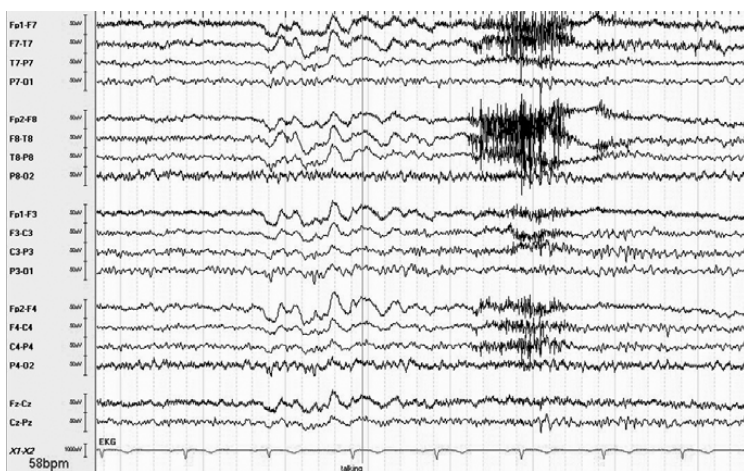


Figure 12.13 Twenty-eight-year-old woman with anxious spells, diffuse rhythmic glossopharyngeal artifact while talking.

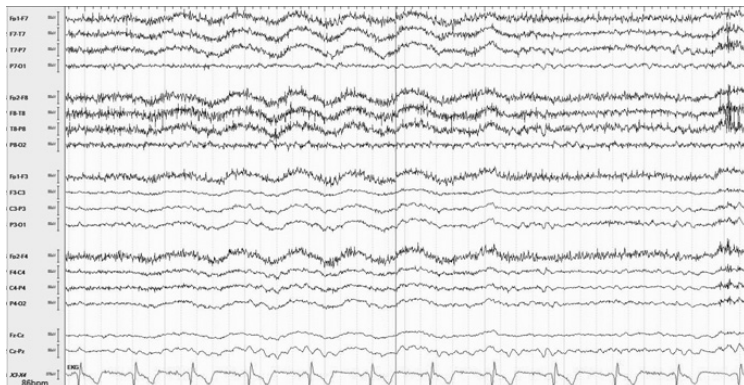


Figure 12.14 Twenty-eight-year-old woman with anxious spells, diffuse low-amplitude rhythmic artifact with sipping through a straw.

Figure 12.15 shows diffuse periodic EKG artifact that is best appreciated in the posterior channels and corresponds to the QRS complex on the EKG channel [4].

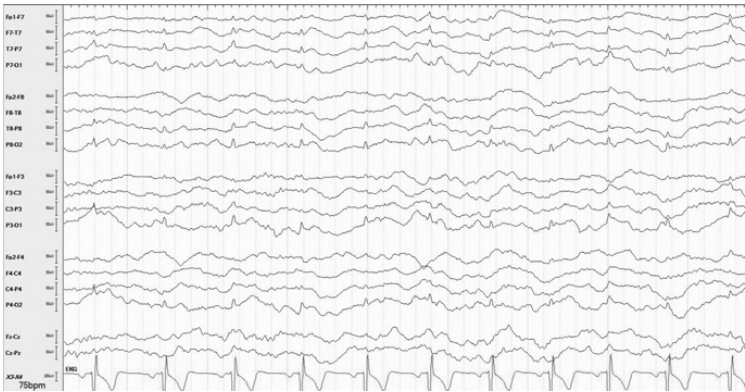


Figure 12.15 Fifty-seven-year-old man with encephalopathy, generalized periodic EKG artifact corresponding to the QRS complex on EKG channel.

Significance

1. This may be confused with periodic discharges, it is identified using the EKG channel.
2. Pacemaker may result in periodic artifacts that just precede the QRS complex.
3. Pulse artifact occurs if an electrode is placed over a pulsating vessel, it is a rhythmic or periodic artifact that follows the QRS.

Cardioballistogram artifact: This is a less commonly seen artifact that results from the rocking motion of the head and neck with forceful heart beats, it mimics slow generalized rhythmic delta activity (GRDA) but also corresponds with the QRS complex. Its duration is usually the same as the inter-QRS interval [3].

Myogenic artifact

Facial Muscles: Typically, myogenic (muscle) artifact is high amplitude, fast (>30 Hz), and spiky. It is most common in the frontal (frontalis), temporal (temporalis), and occipital (occipitalis) regions with the midline channels (Fz, Cz, and Pz) typically free from it, allowing the reader to best observe the underlying cerebral activity in these channels. It is reduced with relaxation, reassurance, and sedation. Neuromuscular blockade in intubated patients also decreases this artifact. Figure 12.16 shows right temporalis muscle artifact [1,2].

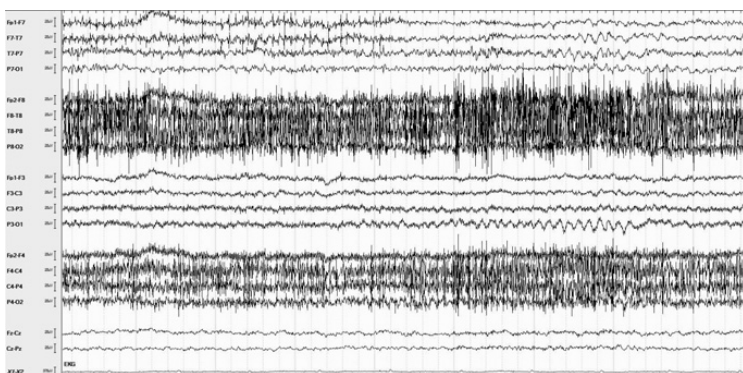


Figure 12.16 Thirty-eight-year-old man with alcohol intoxication, right temporalis muscle artifact.

Significance: Using a high-sensitivity or a high-frequency filter (70 Hz) may cause myogenic artifact to mimic beta activity. True beta, unlike muscle artifact is always seen in the central channels.

Chewing: Chewing results in a prominent generalized periodic burst of myogenic artifact that spares the midline channels. Figure 12.17 shows chewing artifact.

Significance: may resemble a generalized tonic clonic seizure [5].

Myoclonus: Cerebral myoclonus (such as postanoxic) appears as a generalized burst of spike or polyspikes (best seen in central channels) associated with diffuse myogenic artifact [6]. It may be stimulus induced. Figure 12.18 shows myoclonic artifact.

Tremors: Head tremor results in a rhythmic appearing artifact best seen in the posterior channels, while palatal tremor causes a diffuse spike wave-like artifact that is best seen with nasopharyngeal electrodes. These can be confused with electrographic seizures. Figure 12.19 shows head tremor artifact [7].

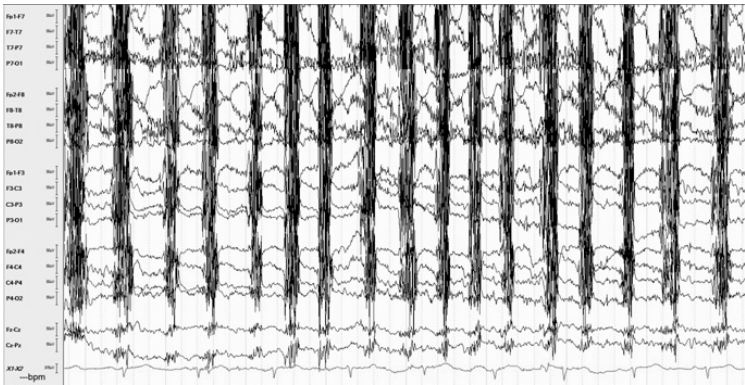


Figure 12.17 Thirty-five-year-old man with shaking spells, diffuse chewing artifact.



Figure 12.18 Sixty-seven-year-old man with cerebral myoclonus, myoclonic discharges best appreciated in the central channels.

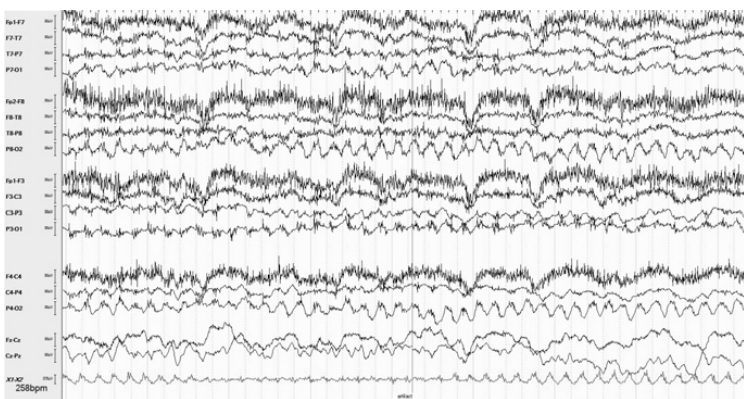


Figure 12.19 Seventy-two-year-old man with head tremor, rhythmic occipital-predominant tremor artifact.

Chapter Summary

1. Artifacts are noncerebral waveforms that may mimic or obscure cerebral activity.
2. External or nonphysiologic artifacts are produced outside the body, while internal or physiologic artifacts are produced by organs other than the brain.
3. Electrode artifacts may have a spiky, periodic, or rhythmic appearance. Characteristically, it is limited to the involved electrode with no field.
4. Sweat artifact may involve multiple channels and may be confused with lateral eye movements or GRDA.
5. Eye movement and glossopharyngeal artifact may mimic frontally predominant GRDA.
6. EKG artifact may be confused with periodic discharges. Characteristically, it corresponds to the QRS complexes.
7. Ventilatory artifact may be confused with bursts of cerebral activity. Characteristically, it corresponds to the respiratory rate.
8. Head tremor presents as occipital-predominant rhythmic artifact.
9. Maneuvers and devices such as bed-percussion, CRRT, ECMO, CPR, and even brushing teeth may lead to ictal appearing rhythmic artifacts.
10. Discharges associated with cortical myoclonus are best appreciated in the central channels, as these are relatively free of muscle artifact.

11. Chewing artifact may electrographically mimic a generalized tonic clonic seizure.
12. Close observation of the patient and the surrounding equipment either in person or on video is the key to diagnosing the cause of the artifact.

References

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